

Public Economics

Lecture 1 · Why Study Public Finance?

Shushanik Margaryan

Winter Term

WU Vienna University of Economics and Business

- This course provides a foundational introduction to public economics
- We meet on Tuesdays from 2pm-6pm
- There will be a break from 3:45-4:15
- Attendance is compulsory. A max of 2 missed sessions is permitted

- A maximum of 100 points can therefore be achieved in total.
 - 3 homework assignments: $3 \times 6 \text{ points} = 18$
 - A midterm and an endterm exam, each worth of 30 points
 - Report (1–3 pages) on a selected research paper in Focus Area 2 = 12 points.
- A min of one point needs to be achieved in each subtask to receive a passing grade.
- Grading scale:
 - < 60 points: Unsatisfactory; 60 points or above: Sufficient; 70 points or above: Satisfactory; 80 points or above: Good; 90 points or above: Excellent;

- PDF document; your name in the file name *and* the header!
- 1–3 pages; more pages does not mean a better score
- English or German
- Inclusive language
- Coherent phrasing, a clear thread throughout, correct spelling, proper citation
- No cover page, no table of contents, ...

- Submit via Canvas — once submitted, it is final
- Submissions are checked for plagiarism! If the text fails this check, you receive 0 points, and the assessment *cannot* be repeated!
- Even translating sentences from the paper into another language counts as plagiarism!

- **Part 1:** Summary of the paper: A clear and correct description of what the paper is about, Research question, objectives, methods, results,... (max. 1 page!)
- **Part 2:** Discuss which aspect of the paper was most interesting to you. What was particularly well done in the paper? Why are you convinced of this?
- **Part 3:** What is missing in the paper? Describe what you think is underdeveloped in the paper, why this aspect would be important, and how you would improve/supplement it. Stick to the context of the present paper and its objective; do not pose a new research question! Refer to the course content and previous texts.

Working Independently

- Work as independently as possible: formulate results, methodology, and the research question in your *own* words!
- Dare to question and critique the papers — we want you to engage with them critically and independently.

A critical use of AI is ok, but

- AI-based tools analyze complex studies only inadequately.
- We discuss the papers in class — you benefit most if you've already read them yourself.
- The papers are exam-relevant, so doing it yourself saves you time in the end.

Introduction

- What is **public finance**, and what are the key questions the field addresses?
- What are the key facts about the size and growth of government, and the distribution of taxes and spending?
- How do the questions of public finance show up in a concrete policy debate — the Covid-19 pandemic?

Introduction

A Motivating Case: Covid-19

1.1 The Four Questions of Public Finance

1.2 Facts on Government in Austria

A Motivating Case: Covid-19

2020: A Natural Experiment in the Role of Government

- March 16, 2020: Austria is among the first countries in Europe to enter a full nationwide lockdown.
- *Kurzarbeit* (short-time work) is massively expanded to keep workers attached to their employers.
- A series of relief instruments follows: *Fixkostenzuschuss*, *Ausfallsbonus*, the Corona-Hilfsfonds.
- February 2022: Austria becomes the *first EU country* to legislate a general Covid-19 vaccination mandate.

Every one of these decisions is a public finance decision.

Source: Covid-19-Maßnahmengesetz, BGBl. I Nr. 12/2020; SN.at, “16. März 2020: Der Tag, ab dem Österreich stillstand”; Impfpflichtgesetz, BGBl. I Nr. 4/2022.

1.1 The Four Questions of Public Finance

What Is Public Finance?

Public finance

The study of the role of **government** in the economy.

The four questions of public finance:

1. **When** should the government intervene in the economy?
2. **How** might the government intervene?
3. **What is the effect** of those interventions on economic outcomes?
4. **Why** do governments choose to intervene in the way that they do?

Q1: When Should the Government Intervene?

Baseline: in a competitive market, the equilibrium (supply = demand) maximizes the *social efficiency* of trade.

- Every trade valued by both sides happens; no valuable trade is missed.
- So the free market is the benchmark — government intervention needs a *justification*.

Two justifications for intervention

1. **Market failure** — the market outcome is not efficient
2. **Redistribution** — society wants a different *distribution* of the outcome (even if it is efficient)

Market Failure

Market failure occurs when a free market fails to allocate resources efficiently.

- Examples of market failures?

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- Examples of market failures?
- Health insurance *looks* like a textbook competitive market: many buyers, many sellers.
- But: if I skip a flu shot, I raise *my* risk of getting the flu — and the risk that I infect *you*.
- I only weigh *my own* costs and benefits when deciding whether to insure/vaccinate; I ignore the cost I impose on others.
- The same logic applies to vaccination generally: below some coverage threshold, *herd immunity* breaks down and outbreaks become possible again — private vaccination choices under-provide protection relative to the social optimum.

⇒ a **negative externality**: private decisions $\neq$ socially efficient decisions.

A Second Market Failure: Adverse Selection

Health insurance also illustrates a completely different reason markets can fail: **asymmetric information**.

- An insurer cannot fully observe an applicant's health risk — but the applicant often has a much better idea of their own risk.
- If insurers cannot price policies by individual risk, they must charge everyone the *average* price — too high for the healthy, too low for the sick.
- Healthy people opt out; the risk pool gets sicker; the average price rises further; more healthy people opt out . . .

Adverse selection

Left unregulated, this spiral can shrink — or entirely unravel — an insurance market, even though every participant would be better off with insurance available. This is exactly what happened, pre-reform, in the individual (“nongroup”) US health insurance market: people with pre-existing conditions were often unable to buy coverage at any price.

The shifting of resources from some groups in society to others.

- Think of the economy as a **pie**: efficiency determines its *size*.
- Redistribution reshuffles different groups' shares of the pie
- Why? Society may value €1 more to a poor household than to a rich one.

The equity–efficiency trade-off

Redistributive policies can change economic agents' behavior (e.g., work incentives), which can *shrink the pie*

Q2: How Might the Government Intervene?

Four broad policy tools:

1. **Tax or subsidize**
2. **Restrict or mandate**
3. **Public provision** (the government produces the good itself)
4. **Public financing of private provision** (government pays; private firms produce)

The Four Tools, With Austrian Examples

Tool	Austrian example
Tax/subsidize	<i>Pendlerpauschale</i> (commuter tax deduction); CO ₂ pricing under the eco-social tax reform
Mandate/restrict	Covid-19 vaccination mandate (2022, repealed 2023)
Public provision	ÖGK public health insurance; public hospitals (Gemeinde-spital network)
Public financing of private provision	Social-insurance reimbursement of private physicians and private hospitals

The Four Tools, All at Once: US Health Insurance

A single country's health insurance system can illustrate *all four* tools simultaneously — exactly why it's such a useful running example:

Tool	US health insurance example
Tax/subsidize	Employer-provided coverage is exempt from income and payroll tax
Mandate/restrict	Obamacare's individual and large-employer coverage mandates (2014–2019)
Public provision	Medicare (age 65+), run directly by the federal government
Public financing of private provision	Medicaid and Obamacare exchange subsidies, channeled through private providers and insurers

As of 2023: 37% of the US population is covered by government insurance, 55% by private insurance, and 8% remain uninsured.

Q3: What Are the Effects of Interventions?

Direct effects

What would happen *if people did not change their behavior*.

Indirect effects

What happens *because* people change their behavior in response.

Both matter for evaluating *any* government intervention

Worked Example: Does Health Insurance Coverage Matter?

Direct effect (naive): give someone insurance $\Rightarrow$ they get more health care $\Rightarrow$ better health. Simple — but comparing the insured to the uninsured directly is misleading: people who get insured differ from those who don't in ways that *also* affect health.

The Oregon Health Insurance Experiment (2008)

Oregon had a limited number of Medicaid slots, so it used a **lottery** to allocate them among 90,000 applicants: 30,000 “winners” got the chance to apply.

Winning the lottery isn't the same as *getting* insured — not all winners enrolled, and some losers got covered another way. Finkelstein et al. (2012) therefore report two effects: the **ITT** (intention-to-treat: simply comparing winners vs. losers) and the **LATE** (scaling the ITT up by $1/0.29$, the actual gap in coverage rates the lottery created) — the *indirect*, causal effect of insurance itself.

Worked Example: What the Oregon Experiment Found

Because the lottery was random, any difference between winners and losers can be attributed to insurance coverage itself — not to who chooses to get insured.

- **Higher health care utilization** — more primary, preventive, *and* hospital care.
- **Lower financial strain** — less medical debt, fewer bills sent to collections.
- **Better self-reported physical and mental health.**

This is Q3 in action: the *direct* effect (more people covered) is only half the story — the *indirect* effects (how covered people actually behave, and what that does to their health and finances) are what a randomized design lets us measure credibly.

Source: Finkelstein et al. (2012), *QJE* 127(3).

Q4: Normative vs. Positive Public Economics

Normative

Analysis of how things *should* be.

Should the government have mandated Covid-19 vaccination? *How high* should the top income tax rate be?

Positive

Analysis of how things *really are*.

Did the Impfpflicht actually raise vaccination rates? *Does* Kurzarbeit actually prevent layoffs, or mostly subsidize jobs that would have survived anyway?

Positive analysis (mostly empirical) is the necessary first step before normative analysis (mostly theoretical) can be completed.

Worked Example: A Live Normative Debate

The US debate over health reform is a clean illustration of *normative* public economics — reasonable people agreeing on the facts but disagreeing on values and trade-offs.

“Medicare for All”

Universal coverage, low copays, funded by taxes. Covers everyone and gives government more leverage to control costs — but requires a huge structural shift away from employer-based coverage.

“Improve Obamacare”

Keep the existing mixed system; patch its gaps (more states expanding Medicaid, richer exchange subsidies). More politically feasible, but preserves a system where cost control is weaker and workers bear coverage costs regardless of income.

Positive economics can tell us what each path would likely cost and who would gain or lose. It cannot tell us which trade-off society *should* choose — that is a normative, political judgment.

1.2 Facts on Government in Austria

Why Look at the Facts?

- Public finance is compelling because government is *everywhere* in economic life.
- We now look at six facts about Austria: size, decentralization, deficits/debt, spending composition, revenue composition, and regulation.
- Where useful, we place Austria in its EU/OECD context

A Methodological Note: What Counts as “Government”?

Measuring the “size of government” first requires deciding *what belongs to* the state sector.

- Under the European System of Accounts (ESVG 2010), Austria’s **Staatssektor** (S.13) comprises **4,973** institutional units across four subsectors: Bund (410), Länder (357), Gemeinden (4,156), and Sozialversicherungsträger (50).
- Classification depends on *function and market/non-market status*, not ownership: a state-owned firm covering most of its costs from market sales (e.g. ÖBB-Personenverkehr, ORF) can fall *outside* the state sector, while a nominally private, state-controlled nonprofit can fall *inside* it.
- **General (“unechte”) Staatsquoten** — like spending/GDP, on the next slide — compare a numerator that is *not* a subset of the denominator, which can overstate government’s real economic weight.
- **Specific (“echte”) Staatsquoten** compare one budget line to the whole budget (e.g. health’s share of total spending) and are more directly interpretable.

Source: Kostal, T., WU Vienna, *Finanzwissenschaft: Öffentliche Finanzen und Budgetpolitik*, lecture slides (Einheit 2: “Erfassung und Messung der Staatstätigkeit”), based on ESVG 2010 and Statistik Austria.

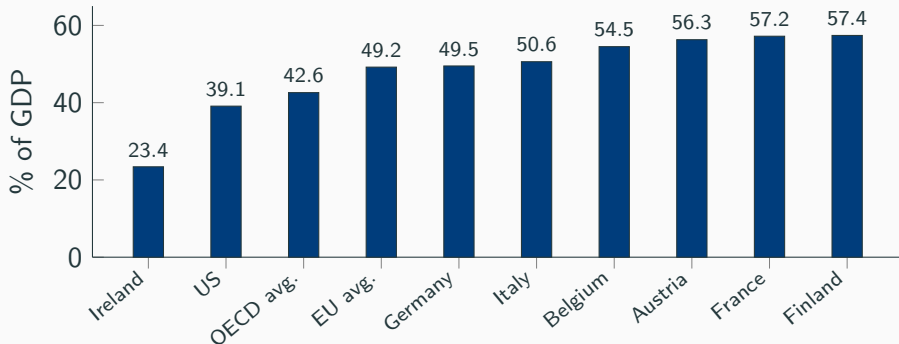
The Size of Government Has Grown — Everywhere

- Government spending has expanded across Western Europe since 1960, driven above all by the build-out of the welfare state.
- The pace has varied enormously: Greece's government share roughly tripled since 1960; Sweden's peaked near two-thirds of GDP in the early 1990s before falling back.
- Austria's own **Staatsquote** sits well above the OECD average today (next slide).

Discussion

What explains the twentieth-century growth of government spending across almost every developed country?

Government Spending Across the EU — and the US



Why Does Government Spending Grow? Four Explanations

No single theory fully explains it — but four recurring arguments:

1. **Wagner's Law:** rising incomes raise the income elasticity of demand for public goods (education, health, infrastructure) above that for private goods — government's share of GDP grows secularly with development.
2. **Displacement effect** (Peacock & Wiseman): crises (wars, pandemics) push taxation/spending to levels citizens would not otherwise accept, and spending rarely falls all the way back — a ratchet.
3. **Baumol's cost disease:** productivity grows more slowly in labor-intensive public services than in the rest of the economy, but public-sector wages must keep pace — raising government's relative cost over time.
4. **Public choice / rent-seeking:** politicians face electoral incentives to expand spending; interest groups lobby for targeted privileges ("distributional coalitions").

Austria's own Covid-19 spending surge (recall the motivating case) illustrates the displacement effect.

Source: Kostal, T., WU Vienna, *Finanzwissenschaft* lecture slides (Einheit 2), citing Wagner; Peacock & Wiseman; Baumol; the public-choice literature.

Austria's federalism

- **Bund** (federal): spends about 54%
- **Länder**: spends about 30%
- **Gemeinden** (municipalities): spends about 16%

A Fourth Player: Sozialversicherung

The previous slide's Bund/Länder/Gemeinden split leaves out a fourth, separately-run subsector: *Sozialversicherung* (social insurance).

Expenditure by level of government (2024)

Bund **48%** (€176.1bn) · Sozialversicherung **25%** (€93.7bn) · Länder **14%** (€52.2bn, excl. Wien) · Gemeinden **13%** (€46.7bn, incl. Wien) — total €272.7bn (55.2% of GDP).

This more precise, sourced breakdown updates the round figures on the previous slide.

Source: Kostal, T., WU Vienna, *Finanzwissenschaft* lecture slides (Einheit 2), citing Statistik Austria, Staatsausgaben nach Teilsektoren (2024).

Why So Centralized? A Political-Economy View

- Redistribution is hard to sustain at the local level: if a *Land* or *Gemeinde* taxes high earners heavily on its own, they can simply move to a lower-tax neighbor.
- Centralizing revenue at the **Bund** is what makes Austria's *Finanzausgleich* possible — transferring resources from richer to poorer regions in a way a fully decentralized system could not.

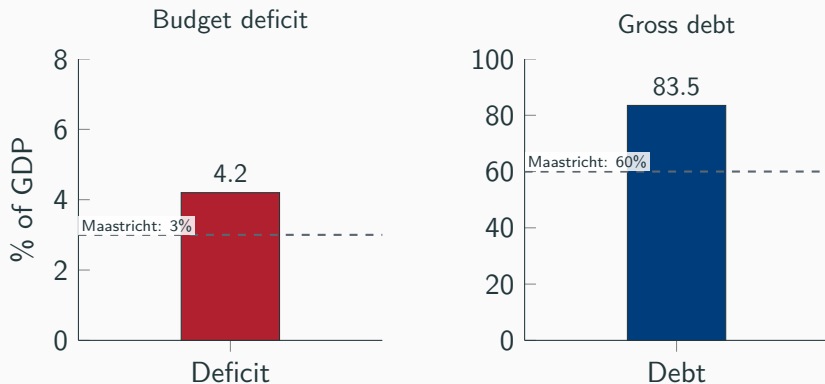
Definitions

Deficit = spending — revenue in a given year.

Debt = accumulated past deficits.

- The EU's **Maastricht rules** cap the deficit at 3% of GDP and gross debt at 60% of GDP for member states.

Austria vs. the Maastricht Rules (2025)



Source: Statistik Austria

Austria's Revenues and Expenditures Over Time

Discussion: Deficits and Debt

- What are the costs of running larger deficits and carrying more debt?
- Why might it make sense to let deficits rise in *some* years (recessions, pandemics) even under fixed fiscal rules?

Austria's Government Expenditure, by Function

Where Does the Revenue Come From?

Across many advanced economies, tax structures have shifted over recent decades — away from taxing corporate profits, toward taxing labor income and consumption.

Discussion

What are the implications of shifting the tax burden from businesses toward workers' earnings?

Austria's Tax Mix: Heavy on Labor

- VAT (*Umsatzsteuer*): $\approx 26.2\%$ (2022)
- Wage tax (*Lohnsteuer*): $\approx 58\%$ (2022)
- Capital tax (*Kapital*): $\approx 16\%$ (2022)

Austria taxes **labor** more heavily than many other EU country.

Source: AK

Austria's Own Income-Distribution Data

- Austria has its own “Distributional National Accounts” (DINA) research, in the tradition of Piketty–Saez–Zucman’s work for the US and France.
- Jestl & List (2020) construct such series for Austria, 2004–2016.

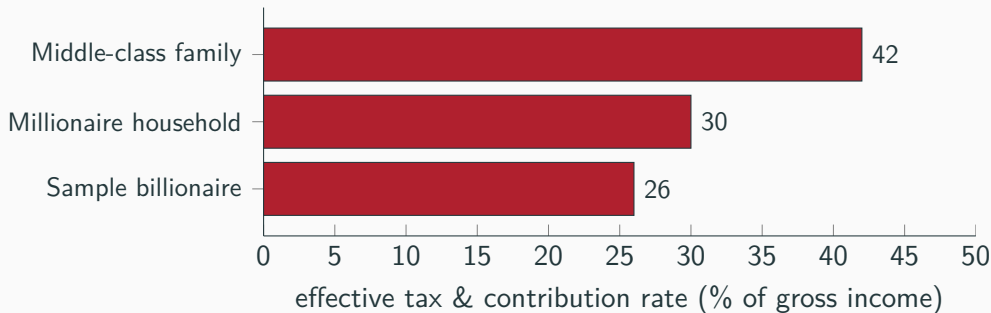
Pre-Tax national income per capita

Average annual real growth rates

Gini coefficient– pre-tax income and post-tax income, 2004-2016

Comparison with France and USA, top 10%

Austria's Tax System, Top to Bottom



The statutory top income-tax rate is 55%—but a billionaire's *income* is mostly business profit and capital gains, taxed on different (lower, flatter) terms than wages.

Source: Momentum Institut, "Problem Steuerstruktur: Arbeit hoch besteuert, Vermögen niedrig,"

Policy Brief 2022.01; "Niedrigsteuerland für Superreiche."

Thank you!

Questions and discussion